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Ten-Word List Performance in Healthy Athletes and Athletes at 3-to-5 Days Following Concussion

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Abstract

Objective: Although the word recall component of the Sport Concussion Assessment Tool (SCAT) was updated from 5-to-10 words, its clinical utility across athletic populations remains unknown. The purpose of this study was to provide normative data on the SCAT-5 10-word test and examine its discriminative ability between healthy university level athletes and those with sport-related concussion (SRC). **Design:** Cross-sectional. **Setting:** University. **Participants:** Three hundred sixty-three (363, M = 220, F = 153) healthy athletes and 49 (M = 30, F = 19) athletes following SRC. **Independent Variable:** Healthy athletes measured preseason were compared with athletes following an SRC (median = 4.0 days postinjury). **Main Outcome Measure:** Ten-word list performance from the SCAT-5. **Results:** There were no significant differences in either immediate ($P = 0.228$, Cohen's D = 0.18) or delayed ($P = 0.908$, Cohen's D = 0.02) recall tests between athletes with SRC and healthy athletes; the null findings extended to both the male and female subset comparisons. Among healthy athletes, females outperformed males on both immediate (mean difference = 1.0, $P < 0.001$) and delayed (mean difference = 0.5, $P = 0.006$) recall tasks. **Conclusion:** Concussion does not seem to impact immediate or delayed recall of the 10-word list in the subacute period following injury. Practitioners should be mindful of sex differences and the time point of administration following injury.

Key Words: head injuries/concussion, female athlete, statistics, cognition, memory, sport concussion assessment tool

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INTRODUCTION

The Sport Concussion Assessment Tool (SCAT) was first published in 2004 by the Concussion in Sport Group (CISG)[1] through amalgamating previously established measures including the Postconcussion Symptom Scale,[2] Standardized Assessment of Concussion (SAC),[3] and the Balance Error Scoring System.[4] Since its inception, there have been subsequent revisions (SCAT-2,[5] SCAT-3,[6] and SCAT-5[7]), which have led to the recognition of the SCAT as the most widely used standardized tool to assist in the diagnosis, management, and prognosis of individuals with concussion.[8]

Despite its ubiquity in the sport context, evidence of the clinical utility for the cognitive components of the SCAT (ie, orientation, concentration, and immediate and delayed recall) is lacking. There are few studies that have examined the clinical utility of the full SCAT. Putukian et al[9] found only symptoms and modified balance error scoring system (mBESS) scores significantly differed from baseline to within 24 hours postinjury using the SCAT-2 in a cohort of collegiate athletes. Extending beyond the 24 hour timeframe, Downey et al[10] found that only the symptoms from the SCAT-3 significantly differed between healthy athletes and athletes at 3-to-5 days following a sport-related concussion (SRC). It has been noted that commonly used cognitive tests may best be characterized not by deficits, but by a lack of otherwise commonly observed practice effects.[9,11] However, interpretation of injury-related deficits without a preinjury (ie, baseline) score remains common practice for many practitioners in the sports medicine and community settings. Thus, understanding the clinical utility of SCAT scores for those without a preinjury evaluation is essential.

A systematic review by Echemendia et al[12] was conducted with the intention of providing information toward improving the sensitivity and/or specificity of the SCAT-3. They found that the immediate recall subcomponent suffered from ceiling effects, potentially limiting the utility of the instrument.[12] The authors suggested that increasing the difficulty of the task (eg, increasing the number of words per trial) may help better detect deficits in verbal learning and memory.[12] The SCAT-5 adopted this recommendation and provided an option to choose a 10-word list instead of a 5-word list for the verbal learning and memory task.[7] In view of this, a recent study in professional hockey

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players examining the diagnostic utility of the SCAT-5 within the first 24 hours of injury found a significant decline in performance of both immediate and delayed recall 10-word list tests.[13] However, other studies that have evaluated the 10-word list test in athletes have been limited to healthy samples.[14–16] Thus, it is still unknown whether there is utility in the 10-word list test for discriminating concussed from healthy athletes beyond professional hockey and whether differences exist between male and female athletes. It is also unclear if the clinical utility of the 10-word list test extends into the subacute period (eg, 3-5 days) for those without a preinjury evaluation; a timeframe where clinical visits, including those to specialized sport and exercise medicine clinics, frequently occur for the assessment of suspected injuries.

Therefore, the purpose of this study was to examine the ability of the 10-word list verbal learning and memory task from the SCAT-5 to discriminate healthy athletes from those with SRC at 3 to 5 days postinjury. Given the lack of sex-specific data in SRC, we also examined potential differences between male and female athletes.

METHODS

Participants

Participant enrollment took place from August 2017 to February 2019 as part of a multiyear study conducted at a single academic institution. Enrollment information for healthy athletes can be seen in Figure 1. Six hundred thirteen healthy athletes spanning 14 sports were baselined before the start of their competitive season, from which 430 athletes agreed to enroll in the study. From the 430 enrolled athletes, 363 healthy baseline athletes (n = 210 male, n = 153 female) were included in the study. Athletes with SRC were contacted for study participation following their diagnosis. The diagnosis was made by a physician with specialization in sport and exercise medicine in accordance with the CISG guidelines[8] and included any of the following: observed signs, subjective reports, and neurologic and cognitive dysfunction associated with either a direct blow to the head, face, neck, or elsewhere on the body with an impulsive force transmitted to the head. Similarly, medical clearance decisions were aligned with consensus guidelines, whereby return to unrestricted activities followed the completion of a graduated exertional strategy.[8] Fifty-four athletes with an acute SRC were enrolled, from which 49 participated (n = 30 male, n = 19 female) (Figure 1). Testing of athletes with SRC occurred within 7 days of injury (median = 4, interquartile range = 3-5). Exclusion criteria for all study participants included a history of SRC within 6 months of study participation. All study participants provided written informed consent before enrollment, and all study procedures were in accordance with the Declaration of Helsinki and approved by the Health Sciences Research Ethics Board, University of Toronto (protocol reference # 27 958).

10-Word List Test

The C3 Logix system is an iPad-based application that has modules that assess for neurocognitive ability in several domains, including symptoms, cognitive functioning (SAC)[17] balance, visual abilities, and visuomotor abilities of reaction time, working memory, and information processing.[18,19] The C3Logix application was aligned with the most recent CISG's Sport Concussion Assessment Tool (SCAT-5),[8,12] beginning with a symptom checklist, followed by the administration of the SAC. For immediate recall, the individual administering the test says each of the 10 words at a rate of one word per second and the participant is required to immediately recall as many words as possible. This process is repeated an additional 2 times, for a total of 3 trials and maximum potential score of 30. The delayed recall test is administered in a similar fashion after the SAC and balance testing, except only a single trial is completed for a maximum potential score of 10. Although the focus of this study was to examine 10-word test performance, orientation and concentration scores can be found in **Supplemental Digital Content 1** (see **Supplementary Table 1**, http://links.lww.com/JSM/A274).

Symptoms

Participant symptoms were determined by a 27-item Post-Concussion Symptom Scale using a 7-point Likert rating as part of the C3 Logix application. This was based on the 22-item scale within the SCAT-5, with the addition of 5 symptoms: "ringing in the ears," "numbness or tingling," "sleeping more than usual," "sleeping less than usual," and "difficulty sleeping soundly." The total symptom score was derived by summing the binary presence/absence of each symptom, up to a maximum score of 27. Symptom severity was calculated by summing the rated symptom score for each symptom.

Statistics

Before all analyses, normality was assessed by comparing variable skewness and kurtosis to a random Gaussian noise model (1000 iterations), followed by the derivation of empirical *P* values. Hence, prior to demographic analysis, variables violating normality were transformed through either a 10% winsorization (minor deviations) or rank transformation (major deviations). Differences in characteristic variables and 10-word test variables between healthy athletes and athletes with SRC were quantitated by calculating the mean difference across 1000 bootstrapped resamples, followed by the derivation of Cohen's d effect sizes. Fractional *P* values were identified for each variable by computing the fraction of bootstrapped coefficient values not enclosing zero effect in a 2-tailed framework, corrected at a false discovery rate of 0.05. For categorical variables, the mean difference test was evaluated on the percent of individuals categorized to each outcome, with effect sizes described as odds ratios. Given the difference in age and concussion history between healthy athletes and athletes with SRC (Table 1), a secondary analysis was conducted by using partial regression to adjust for age and concussion history on immediate and delayed recall scores between healthy athletes and athletes with SRC.

Reference ranges for 10-word list scores in healthy athletes found in Table 2 were created according to cut points for "borderline" and "impaired" scores as previously reported[20,21]; borderline impaired scores were characterized as between the mean $-1.3 \times$ the SD and the mean $-2 \times$ SD, whereas impaired scores were characterized as all values below the mean $-2 \times$ SD. To calculate the probability for each test score (immediate and delayed recall tests), a Gaussian cumulative distribution function was used to calculate the probability of having an equal or lesser score for each test

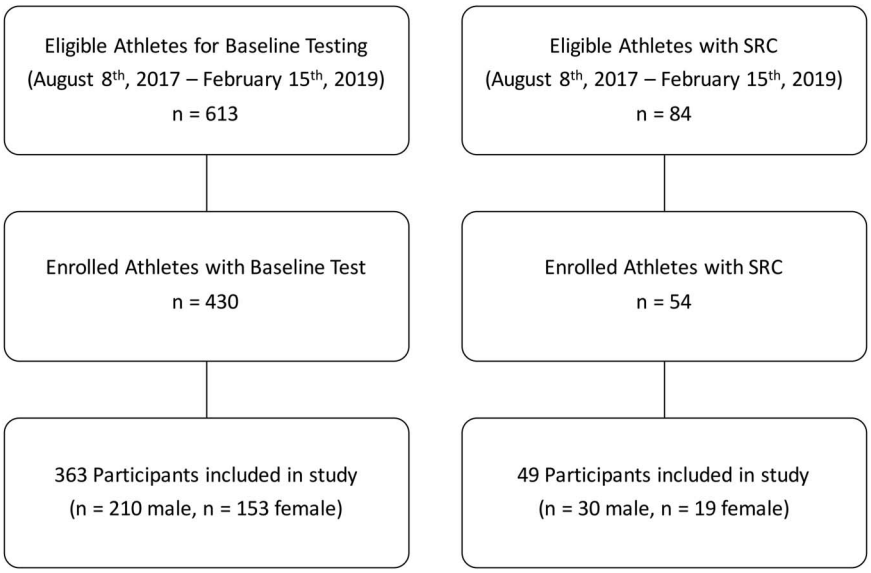

Figure 1. Athlete enrollment and study participant selection. Baseline athletes: 613 athletes were eligible for baseline testing, 430 athletes enrolled. From this number, 33 samples were removed because of repeated baseline tests, 28 were removed as they were subsequently concussed and became eligible for the concussion group, and 6 were removed because of missing data. As a result, the study included 363 healthy athletes (n = 210 male, n = 153 female). SRC athletes: 84 athletes were eligible for study participation following a SRC. From this number, 54 were enrolled in the study. Two athletes were removed because of missing C3 recall data, 2 were removed because of repeated SRCs, and 1 athlete was removed because of missing data. As a result, 49 athletes with SRC were included in the study (n = 30 male, n = 19 female). SRC; sport-related concussion.

TABLE 1. Athlete Characteristics

Variable	SRC (n = 49)	Healthy (n = 363)
Age (yr)	20.9 (19.4-21.9)*	19.3 (18.3-21.2)
Sex (n, % male)	30 (61.2)	210 (57.8)
Concussion history (n, % yes)	27 (55.1)*	113 (31.1)
Years since last concussion	1.6 (1.1-3.2)	2.9 (1.6-4.2)
Total symptoms	12 (7-19)*	2 (0-5)
Symptom severity	29 (11-48)*	3 (0-7)
SAC concentration	4.1 (0.9)	3.8 (1)
SAC orientation (n = 5, %)	44 (89.8)	302 (83.2)
Days from concussion to testing	4 (3-5)	
Days to medical clearance†	35 (17-66)	—
Sport		
Baseball	2 (4.1)	—
Basketball	—	34 (9.4)
Field hockey	2 (4.1)	15 (4.1)
Figure skating	2 (4.1)	—
Football	9 (18.7)	81 (22.3)
Ice hockey	6 (12.2)	41 (11.3)
Lacrosse	1 (2)	39 (10.7)
Mountain biking	2 (4.1)	—
Rowing	1 (2)	—
Rugby	15 (30.6)	51 (14)
Soccer	3 (6.2)	45 (12.4)
Softball	1 (2)	—
Swimming	1 (2)	—
Volleyball	2 (4.1)	26 (7.2)
Water polo	2 (4.1)	31 (8.5)

All values displayed as the median and interquartile range, unless otherwise stated.
** Significantly higher compared with healthy athletes by resampled mean difference testing at a percentile P value corrected at a false discovery rate of 0.05.*
† RTP dates for 43 athletes.
SRC, sport-related concussion; SAC, Standardized Assessment of Concussion.

score. Density plots for immediate and delayed recall (Figures 2 and 3) were derived using a Gaussian smoothing kernel.

All statistical analyses were performed using R (RStudio, version 1.2.1335, Boston, United States).

Results

Participant Characteristics

Participant characteristics can be viewed in Table 1. Athletes with SRC were significantly older than healthy athletes (mean difference = 0.8 years, Cohen's d = 0.47, $P = 0.004$) and were more likely to have had a history of concussion (55.1 vs 31.1%, respectively: OR = 1.8, $P < 0.001$). Compared with healthy athletes, those with SRC reported significantly greater total symptoms (mean difference = 10, Cohen's d = 1.54, $P < 0.001$) and symptom severity (mean difference = 26, Cohen's d = 1.55, $P < 0.001$).

Word Recall in Healthy Athletes

There were no significant differences between healthy athletes with or without a history of concussion on either immediate (21.5 vs 20.9, respectively, Cohen's d = 0.2, $P = 0.078$) or delayed (7.3 vs 7.0, respectively, Cohen's d = 0.18, $P = 0.126$) recall. However, differences between males and females were observed; healthy female athletes displayed significantly higher scores in both tests: immediate recall (21.7 vs 20.7, mean difference = 1, Cohen's d = 0.35, $P = 0.002$) and delayed recall (7.3 vs 6.8, mean difference = 0.5, Cohen's d = 0.29, $P = 0.006$). Normal, borderline-impaired, and impaired reference ranges for recall scores on all healthy athletes can be found in Table 2. In addition, Tables 3 and 4 display the probabilities for each test score for both immediate and delayed recall tests in healthy athletes, stratified by sex.

10-Word List Comparison Between Healthy Athletes and Athletes After Sport-Related Concussion

Statistical results for the 10-word list test between healthy athletes and those with an SRC are described in Table 5, and density plots can be visualized in Figures 2 and 3. In healthy athletes, immediate recall scores ranged from 11 to 30 with only a single (0.3%) male participant achieving a perfect score; scores ranged from 13 to 30 in males and 11 to 28 in females. Delayed recall scores in healthy athletes ranged from 2 to 10, with 26 (7.2%) participants achieving a perfect score; scores ranged from 2 to 10 (4.3% perfect) in males and from 3 to 10 (11.1% perfect) in females. In athletes with SRC, immediate recall ranged from 15 to 30, with a single (2%) female participant achieving a perfect score; scores ranged from 15 to 27 in males and 19 to 30 in females. Delayed recall scores in athletes after SRC ranged from 2 to 10, with 4 (8.2%) athletes achieving a perfect score; scores ranged from 2 to 10 (6.7% perfect) in males and from 3 to 10 (10.5% perfect) in females.

There were no significant differences between healthy athletes and those with an SRC for either immediate or delayed recall tests (Figure 2A, B, respectively). Furthermore, separate subset analyses on male and female participants also yielded no significant differences between healthy athletes and those with an SRC for either immediate recall (Figure 3A, B, respectively) or delayed recall (Figure 3C, D, respectively). Analogous results were observed after adjusting for age and concussion history (see **Supplementary Table 2, Supplemental Digital Content 1**, http://links.lww.com/JSM/A275). When using normative cut-values (Table 2), no athletes were considered impaired on the immediate recall test following SRC, and 10.2% of all athletes were considered borderline-impaired (scores = 15-17). However, when separated by sex, 10% of males (scores = 15-16) and no females (scores = 16-17) presented with borderline-impaired immediate recall scores. For the delayed recall test, 4.1% (scores ≤3) of all athletes, 3.3% (scores ≤3) of male athletes, and 5.3% (scores ≤4) of female athletes had impaired scores following SRC. In addition, after SRC borderline-impaired delayed recall scores were observed for no athletes when evaluated altogether (scores = 4), no male athletes (scores = 4), yet 21.1% (scores = 5) of female athletes.

DISCUSSION

In 2017, the immediate and delayed recall tasks within the SCAT were expanded from 5 to 10 words to alleviate potential ceiling effects and improve test sensitivity.[12] In the intervening years, there have been limited studies reporting the utility of these changes. The purpose of the current study was to evaluate the performance on the 10-word list immediate and delayed recall test in discriminating healthy university athletes from athletes at 3-to-5 days (median = 4 days) following an

TABLE 2. Reference Ranges for 10-Word List Scores in Healthy Athletes

Sample	Normal*	Borderline†	Impaired‡
Immediate recall			
All (n = 363)	≥18 (83.5%)	15-17 (15.1%)	≤14 (1.4%)
Male (n = 210)	≥17 (89.5%)	15-16 (8.6%)	≤14 (1.9%)
Female (n = 153)	≥18 (88.9%)	16-17 (9.8%)	≤15 (1.3%)
Delayed recall			
All (n = 363)	≥5 (90.4%)	4 (6.9%)	≤3 (2.7%)
Male (n = 210)	≥5 (88.1%)	4 (8.1%)	≤3 (3.8%)
Female (n = 153)	≥6 (81.7%)	5 (9.8%)	≤4 (5.2%)

Values presented as the score value and percent —(%).
** Normal is represented as all values greater than the mean − 1.3 × SD.*
† Borderline is represented as all values between the mean − 1.3 × SD & mean − 2 × SD.
‡ Impaired is represented as all values ≤mean − 2 × SD.

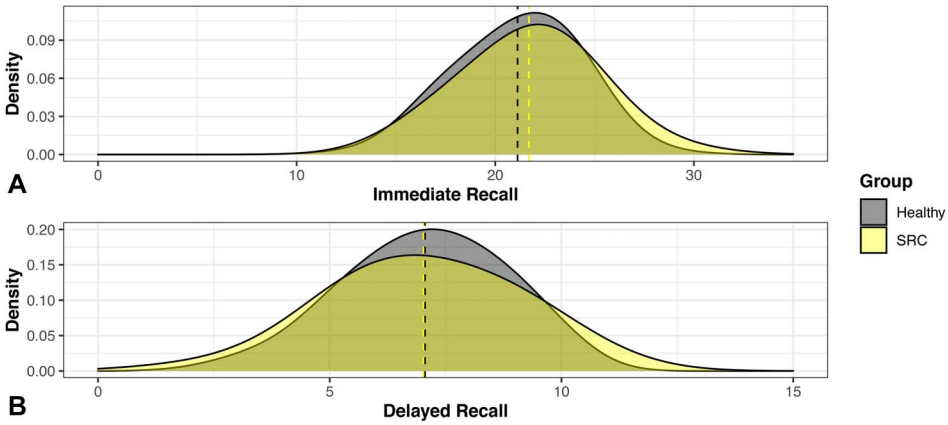

Figure 2. Density plots of immediate and delayed recall in all athletes. Density plots of immediate memory recall (A) and delayed recall (B) in healthy athletes (n = 363, gray) and athletes following sport-related concussion (SRC) (n = 49, yellow) were created using Gaussian kernel estimation. Dotted lines represent group means.

SRC. We also sought to evaluate this comparison separately on male and female athletes. We observed no difference between groups on either task.

Our observed null findings in mean immediate and delayed recall scores between healthy athletes and athletes with SRC extended to male and female comparisons. Our findings are similar to a prior investigation by our group using the 5-word list test used within the same testing time frame.[10] Yet, our results also seem contrary to recent published data in professional hockey players by Bruce et al[13] that observed significant decrements in both immediate and delayed recall after suspected concussion. We believe this is largely because of the timeframe of testing; in the aforementioned study, athletes were sampled within 1 day of injury, whereas in the current study, a median of 4 days elapsed from injury to testing. It is plausible that the 10-word list test is sensitive to the acute, but not subacute effects of concussion, and hence, the current study serves to elucidate the timeframe bounds of utility for the 10-word list test.

Interestingly, we observed a counterintuitive relationship between concussion history and both immediate and delayed recall in participants following SRC; the number of prior reported concussions was associated with improved immediate and delayed recall scores. Although it is unclear why this was observed, there is reason to suggest that procedural familiarity (ie, practice effects) may have contributed. In our sample, athletes with no history of concussion had been administered an average of 1.9 SCAT tests (including the current test), whereas those with a history of concussion had been administered an average of 3.7 SCAT tests. Furthermore, there was a significant correlation between the number of SCAT tests administered and immediate recall within both the baseline ($r = 0.19$, $P < 0.001$) and SRC ($r = 0.44$, $P < 0.001$) groups. This raises the important question of the influence of procedural familiarity on immediate and delayed recall and how this may affect test utility. Future research on the confounding role of procedural familiarity on the 10-word list test is necessary.

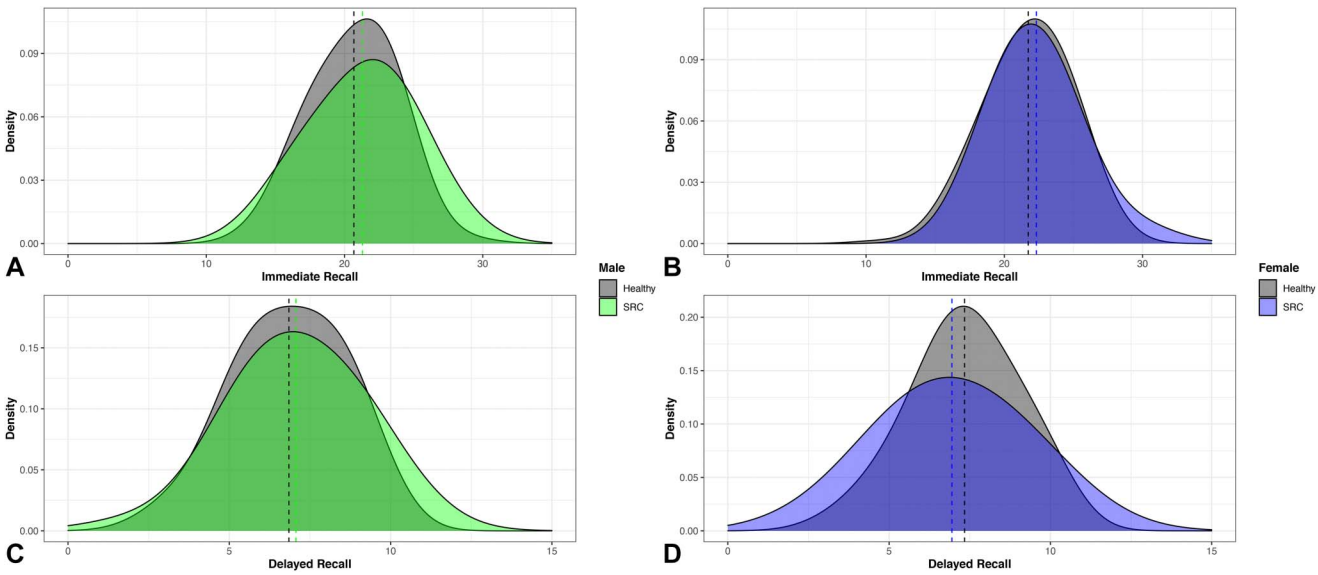

Figure 3. Density plots of immediate and delayed recall in male and female athletes. Density plots of immediate (A, B) and delayed (C, D) recall in healthy athletes (male n = 210, female n = 153, gray) and athletes following sport-related concussion (SRC) (male n = 30, green; female n = 19, purple) were created using Gaussian kernel estimation. Dotted lines represent group means.

TABLE 3. Immediate Recall—Probability by Score for Healthy Athletes

Score	All (n = 363)	Male (n = 210)	Female (n = 153)
11	0	0.1	0
12	0.1	0.2	0.1
13	0.4	0.6	0.2
14	1	1.5	0.5
15	2.3	3.2	1.2
16	4.8	6.4	2.7
17	9	11.5	5.7
18	15.5	19.1	10.6
19	24.5	29.2	18
20	35.8	41.3	28.1
21	48.4	54.2	40.3
22	61.3	66.7	53.6
23	73	77.6	66.4
24	82.6	86.1	77.6
25	89.7	92.1	86.3
26	94.4	95.9	92.3
27	97.2	98	96.1
28	98.7	99.2	98.2
29	99.5	99.7	99.3
30	99.8	99.9	99.7

Data presented as the probability of having an equal or lesser score on the test, calculated using a Gaussian cumulative distribution function.

The results indicate the implementation of the optional 10-word list achieved the desired outcome of reducing perceived ceiling effects of the 5-word list.[7,12,16] Because we observed a Gaussian distribution for both delayed and immediate recall with only a small proportion of perfect test scores, suggests that the 10-word list (immediate and delayed) is not hindered by a ceiling. Indeed, only 0.3% and 7.2% of healthy athletes achieved perfect test scores for immediate and delayed recall, respectively; in athletes following, SRC these values were 2% and 8.2%. Importantly, the test also seems reliable, as our results correlate well with 2 previous studies reporting 10-word list performance assessment in a healthy college population[16] and high-school student athletes.[14] Hence, we believe the 10-word list test to be an improvement over the prior 5-word version list in the SCAT-3, regardless of its inability to discriminate healthy athletes from athletes with an SRC in our sample during the subacute timepoint.

In the current study, female athletes performed significantly better than males on both immediate and delayed recall. This is supported by the findings of Norheim et al[16] and is historically consistent with prior studies using the SCAT-3,[22] as well as computerized cognitive test results.[23] Furthermore, we observed a trend suggesting an increased number of females with borderline delayed recall test scores following SRC compared with males; importantly, this was not observed when males and females were evaluated together, because the normative cut-value is not adjusted to account for the slightly higher performance by females on delayed recall (a score of 4 vs 5 in the female group, Table 2). This, taken together with the growing body of evidence that suggests females have a unique symptom burden and recovery

TABLE 4. Delayed Recall—Probability by Score for Healthy Athletes

Score	All (n = 363)	Male (n = 210)	Female (n = 153)
1	0	0	0
2	0.2	0.3	0.1
3	0.9	1.4	0.4
4	3.7	5.1	2.1
5	11.5	14.5	7.6
6	26.9	31.3	20.6
7	48.7	53.5	41.8
8	70.9	74.6	65.7
9	87.2	89.2	84.5
10	95.7	96.5	94.8

Data presented as the probability of having an equal or lesser score on the test, calculated using a Gaussian cumulative distribution function.

TABLE 5. 10-Word List Tests Comparison Between Healthy Athletes and Those With an SRC

Groups	SRC	Healthy	Mean Difference (95% CI)	*P*	Cohen's D
Immediate recall					
All athletes	21.7 (3.1)	21.1 (3.1)	0.6 (−0.3 to 1.5)	0.228	0.18
Males	21.3 (3.3)	20.7 (3.1)	0.6 (−0.7 to 1.8)	0.320	0.20
Females	22.3 (2.7)	21.7 (3)	0.6 (−0.6 to 1.9)	0.404	0.20
Delayed recall					
All athletes	7 (1.8)	7.0 (1.7)	−0.02 (−0.5 to 0.5)	0.908	0.02
Males	7.1 (1.8)	6.8 (1.7)	0.2 (−0.5 to 0.8)	0.484	0.14
Females	6.9 (1.9)	7.3 (1.6)	−0.4 (−1.3 to 0.5)	0.416	0.22

All values displayed as the mean and SD unless otherwise stated. Differences were quantitated by a resampled mean difference test (1000 iterations). Significance was corrected at a false discovery rate of 0.05.
SRC, sport-related concussion.

trajectory after concussion compared with males[24] and also display differences in postinjury pathophysiology,[25,26] further reiterates the necessity to move away from combined male and female group analyses in concussion research.

The current study must be interpreted within the context of its limitations. Although the sample size was large enough to support a robust and stable statistical analysis, a larger number of concussed athletes would have allowed for subgroup analyses that could have been expanded to include sport type and concussion history within the healthy and SRC comparison. Furthermore, and as alluded to above, a larger sample size would have also allowed for the controlling of test number, as we saw evidence of procedural familiarity on the 10-word list. Finally, it must be remembered that the results of this study cannot be generalized beyond university-level athletes.

The shift from a 5-to-10 word list test seems to have alleviated test ceiling effects as measured in male and female university athletes spanning a variety of sports. However, in the subacute period after injury, athletes with SRC show no differences in immediate or delayed recall test scores compared with healthy athletes. Moreover, females perform better on both tasks compared with males in the healthy sample. Future research using SCAT-5 10-word list recall should be cognizant of the difference in test performance at different phases of recovery following concussion, sex differences, and the potential confounding influence of procedural familiarity.

References

1. McCrory P. Summary and agreement statement of the 2nd international conference on concussion in sport. *Praguebrit J Sport Med*. 2005; 39(supplement 1):i78–i86.
2. Lovell MR, Collins MW. Neuropsychological assessment of the college football player. *J Head Trauma Rehabil*. 1998;13:9–26.
3. McCrea M, Kelly JP, Randolph C, et al. Standardized assessment of concussion (SAC): on-site mental status evaluation of the athlete. *J Head Trauma Rehabil*. 1998;13:27–35.
4. Riemann BL, Guskiewicz KM. Effects of mild head injury on postural stability as measured through clinical balance testing. *J Athl Train*. 2000;35:19–25.
5. McCrory P, Meeuwisse W, Johnston K, et al. Consensus statement on concussion in sport 3rd international conference on concussion in sport held in Zurich, November 2008. *Clin J Sport Med*. 2009;19:185–200.
6. McCrory P, Meeuwisse WH, Aubry M, et al. Consensus statement on concussion in sport: the 4th international conference on concussion in sport held in Zurich, November 2012. *Br J Sports Med*. 2013;47: 250–258.
7. Echemendia RJ, Meeuwisse W, McCrory P, et al. The sport concussion assessment tool 5th edition (SCAT5): background and rationale. *Br J Sports Med*. 2017;51:848–850.
8. McCrory P, Meeuwisse W, Dvorak J, et al. Consensus statement on concussion in sport-the 5th international conference on concussion in sport held in Berlin, October 2016. *Br J Sport Med*. 2017;51:838–847.
9. Putukian M, Echemendia R, Dettwiler-Danspeckgruber A, et al. Prospective clinical assessment using Sideline Concussion Assessment Tool-2 testing in the evaluation of sport-related concussion in college athletes. *Clin J Sport Med*. 2015;25:36–42.
10. Downey RI, Hutchison MG, Comper P. Determining sensitivity and specificity of the Sport Concussion Assessment Tool 3 (SCAT3) components in university athletes. *Brain Inj*. 2018;32:1345–1352.
11. Echemendia RJ, Putukian M, Mackin RS, et al. Neuropsychological test performance prior to and following sports-related mild traumatic brain injury. *Clin J Sport Med*. 2001;11:23–31.
12. Echemendia RJ, Broglio SP, Davis GA, et al. What tests and measures should be added to the SCAT3 and related tests to improve their reliability, sensitivity and/or specificity in sideline concussion diagnosis? A systematic review. *Br J Sports Med*. 2017;51:895–901.
13. Bruce JM, Thelen J, Meeuwisse W, et al. Use of the Sport Concussion Assessment Tool 5 (SCAT5) in professional hockey, part 2: which components differentiate concussed and non-concussed players?. *Br J Sports Med*. 2020. doi: 10.1136/bjsports-2020-102072 [epub ahead of print].
14. Black AM, Miutz LN, KV VW, et al. Baseline performance of high school rugby players on the sport concussion assessment tool 5. *J Athletic Train*. 2020;55:116–123.
15. Tucker R, Brown J, Falvey E, et al. The effect of exercise on baseline SCAT5 performance in male professional Rugby players. *Sports Med Open*. 2020;6:37.
16. Norheim N, Kissinger-Knox A, Cheatham M, et al. Performance of college athletes on the 10-item word list of SCAT5. *BMJ open Sport Exerc Med*. 2018;4:e000412.
17. McCrea M. Standardized mental status testing on the sideline after sport-related concussion. *J Athl Train*. 2001;36:274–279.
18. Bernstein JPK, Calamia M, Pratt J, et al. Assessing the effects of concussion using the C3Logix Test Battery: an exploratory study. *Appl Neuropsychol Adult*. 2019;26:275–282.
19. Simon M, Maerlender A, Metzger K, et al. Reliability and concurrent validity of select C3 Logix test components. *Dev Neuropsychol*. 2017;42:446–459.
20. Wheeler D. *Wechsler Adult Intelligence Scale-Revised, Manual*. New York, NY: Psychological Corporation; 1981.
21. Palmer BW, Boone KB, Lesser IM, et al. Base rates of "impaired" neuropsychological test performance among healthy older adults. *Arch Clin Neuropsychol*. 1998;13:503–511.
22. Chin EY, Nelson LD, Barr WB, et al. Reliability and validity of the sport concussion assessment tool-3 (SCAT3) in high school and collegiate athletes. *Am J Sports Med*. 2016;44:2276–2285.
23. Covassin T, Swanik CB, Sachs M, et al. Sex differences in baseline neuropsychological function and concussion symptoms of collegiate athletes. *Br J Sports Med*. 2006;40:923–927; discussion 927.
24. Iverson GL, Gardner AJ, Terry DP, et al. Predictors of clinical recovery from concussion: a systematic review. *Br J Sports Med*. 2017;51:941–948.
25. Di Battista AP, Churchill N, Rhind SG, et al. The relationship between symptom burden and systemic inflammation differs between male and female athletes following concussion. *BMC Immunol*. 2020;21:11.
26. Hutchison MG, Mainwaring L, Senthinathan A, et al. Psychological and physiological markers of stress in concussed athletes across recovery milestones. *J Head Trauma Rehabil*. 2017;32:E38–e48.